AI ACADEMY / MLE-AI-201

Understanding Bina.az sale listings

From-scratch Decision Tree and Pegasos SVM

Price regression and Premium listing classification

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RESEARCH QUESTION

Two questions from one dataset

Bina.az sale listings contain property details and the asking price.

REGRESSION

How much?

Predict the listing price in AZN.

CLASSIFICATION

Premium or Standard?

Premium means price above the training median: 210,000 AZN.

Goal: implement the algorithms ourselves, then compare them with sklearn reference models.

DATA PREPARATION

Cleaning produces a usable feature set

100,775
raw listings

64,128
cleaned listings

29
model features

We removed repeated listings and unusable rows, then encoded categorical fields.

Leakage check

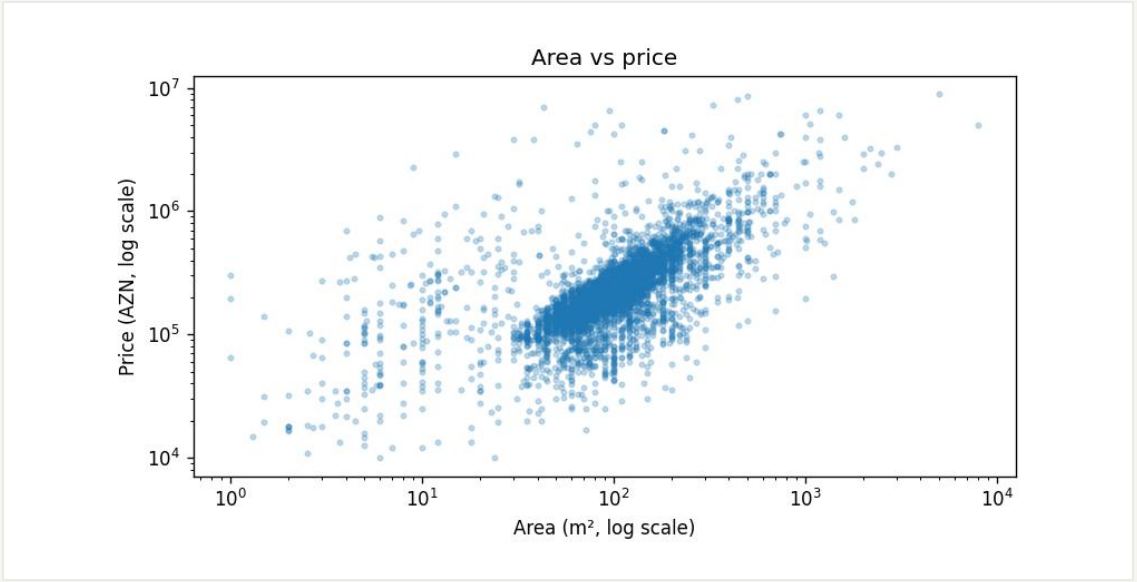
unit_price and total_price reveal the target and therefore cannot be used as predictors.

EXPERIMENTAL SETUP

Train-only decisions protect the test set

TRAIN 44,890 | **VALIDATION 9,619** | **TEST 9,619** | **70% / 15% / 15%**

The Premium threshold and SVM scaling parameters come from training data only.



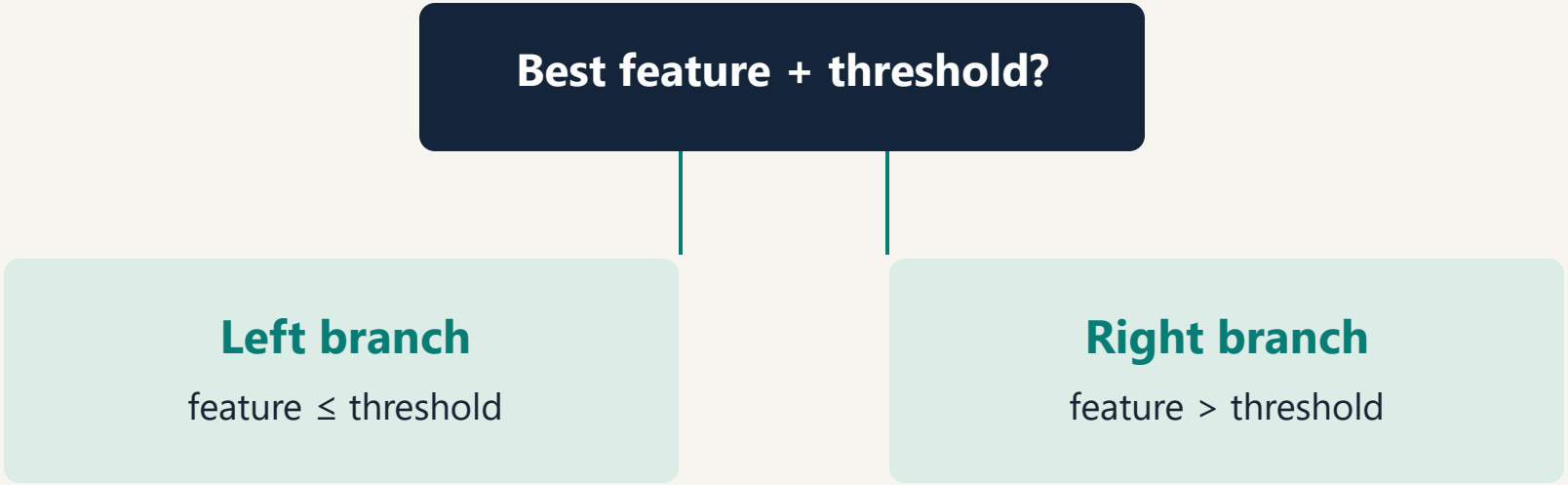
What the data shows

Area rises with price, but listings vary widely at the same area.

This helps explain why price regression remains difficult.

A tree learns one split at a time

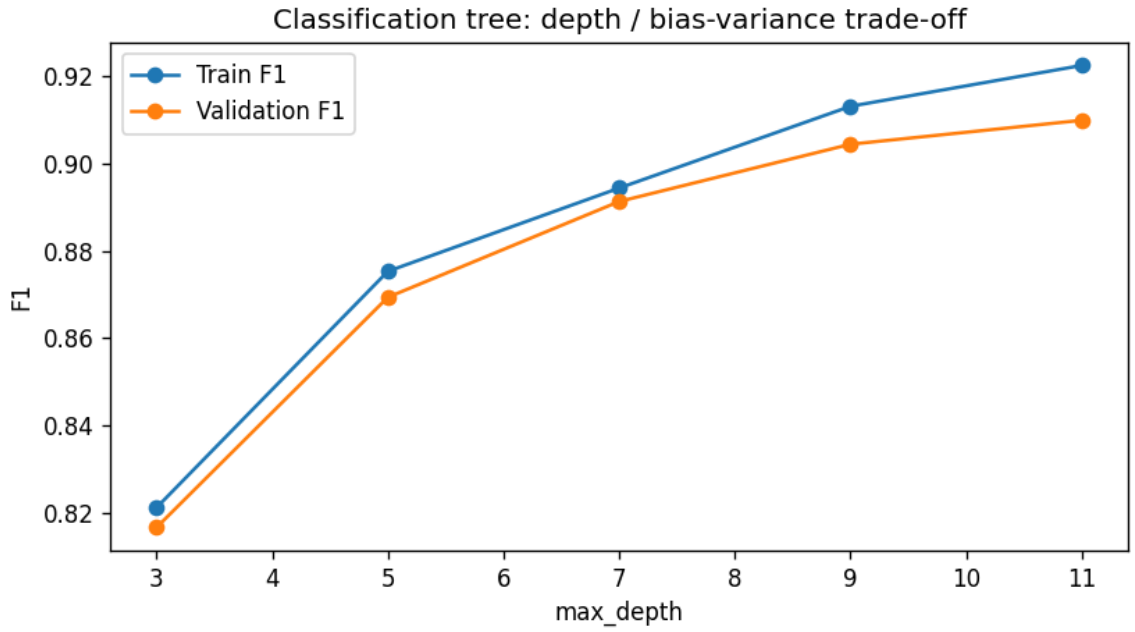
At each node, search feature values for the split with the largest impurity reduction.



Classification: Gini or entropy

Regression: MSE / variance reduction

Our tree closely matches sklearn



Test classification F1

Our Decision Tree	**0.910**
sklearn Decision Tree	**0.910**

ROC-AUC: 0.963 for both models

Deeper trees improved validation F1 within the tested range.

Tree regression: R^2 = 0.592 and MAE = 87,181 AZN.

Pegasos learns a separating margin

The linear SVM separates Premium and Standard listings with a score.

Hinge loss

max(0, 1 − y · score)

Wrong or low-margin predictions receive a penalty.

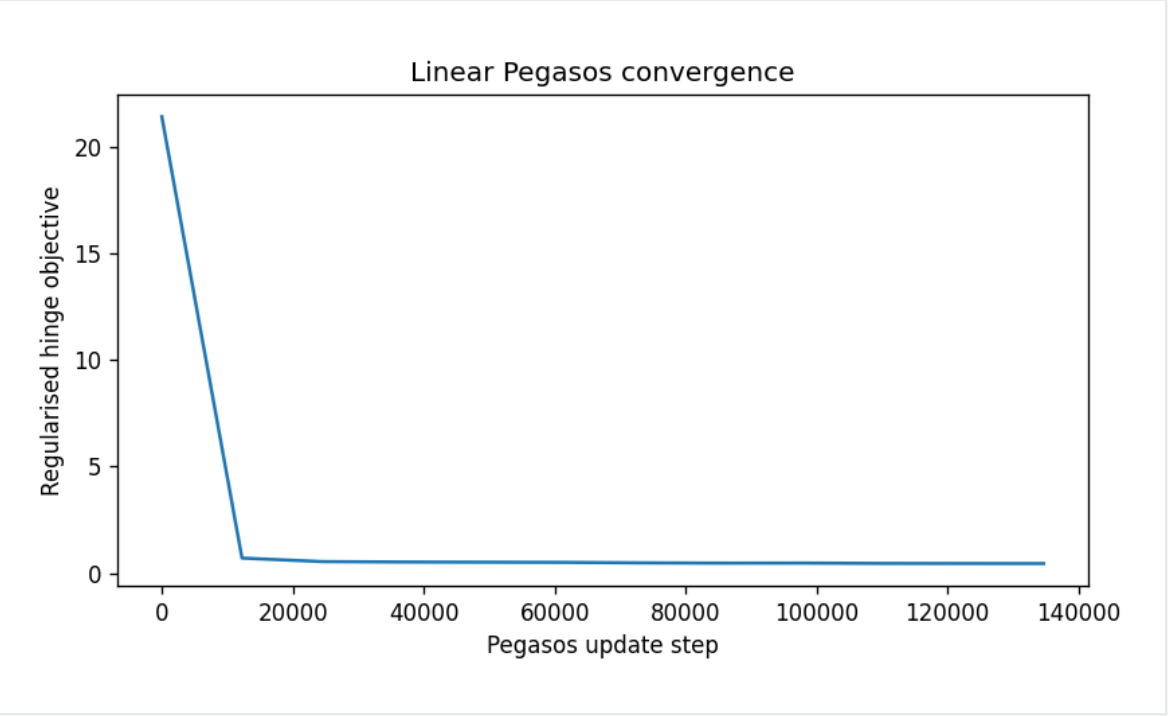
Pegasos update

Repeated weight updates minimize hinge loss with L2 regularization.

λ, epochs and learning schedule control training.

Support vectors are points close to, or inside, the margin.

The nonlinear extension did not improve F1



Test F1

Linear Pegasos	**0.823**
sklearn SGD hinge	**0.826**
RFF + Pegasos	**0.810**

RFF approximates a nonlinear feature map before the same linear SVM.

The objective falls quickly, then stabilizes during training.

Classification is stronger than price regression

Held-out test set / 9,619 listings

Model	F1	ROC-AUC	Comment
Our Tree	0.910	0.963	best F1
sklearn Tree	0.910	0.963	reference
Our Linear Pegasos	0.823	0.908	close to SGD
Our RFF Pegasos	0.810	0.894	lower F1

Regression limit Top 5% price listings: MAE ≈ 773k AZN vs ≈ 49k AZN for the rest.

One pipeline, three primary ownership areas

Samir Hasanov	**Decision Tree from scratch** Integration and tree experiments
Mammadaliyev Ramil	**Pegasos SVM and RFF** Margin, convergence and SVM tests
Elmin Seyidov	**Data preparation and EDA** Leakage checks, splits and metrics

Takeaway: correct implementation and fair evaluation matter more than a high score alone.